

Topic 9: Control Systems

This topic builds on knowledge of transport mechanisms and considers the processes of chemical and nervous coordination. Details of the role of plant growth substances and hormonal control in mammals are included. Osmoregulation in mammals adapted to dry environments is considered. Details of the effect of drugs on the transmission of nerve impulses are studied. Practical skills are developed in the investigation of the effect of gibberellin on germination.

Opportunities for developing mathematical skills within this topic include: making use of appropriate units in calculations; using expressions in decimal and standard form; using ratios, fractions and percentages; interpreting frequency tables and diagrams, bar charts and histograms; understanding simple probability; translating information between graphical, numerical and algebraic forms; plotting variables from experimental data; understanding that $y = mx + c$ represents a linear relationship; determining the intercept of a graph; calculating rate of change from a graph; using the slope of a tangent to a curve as a measure of rate of change. (Please see *Appendix 6: Mathematical skills and exemplifications* for further information.)

Students should:

9.1 Homeostasis

- i Know that homeostasis is the maintenance of a state of dynamic equilibrium.
- ii Understand the importance of maintaining pH, temperature and water potential in the body.
- iii Understand what is meant by negative feedback and positive feedback control.